

MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY



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Frequency as a scarce resource

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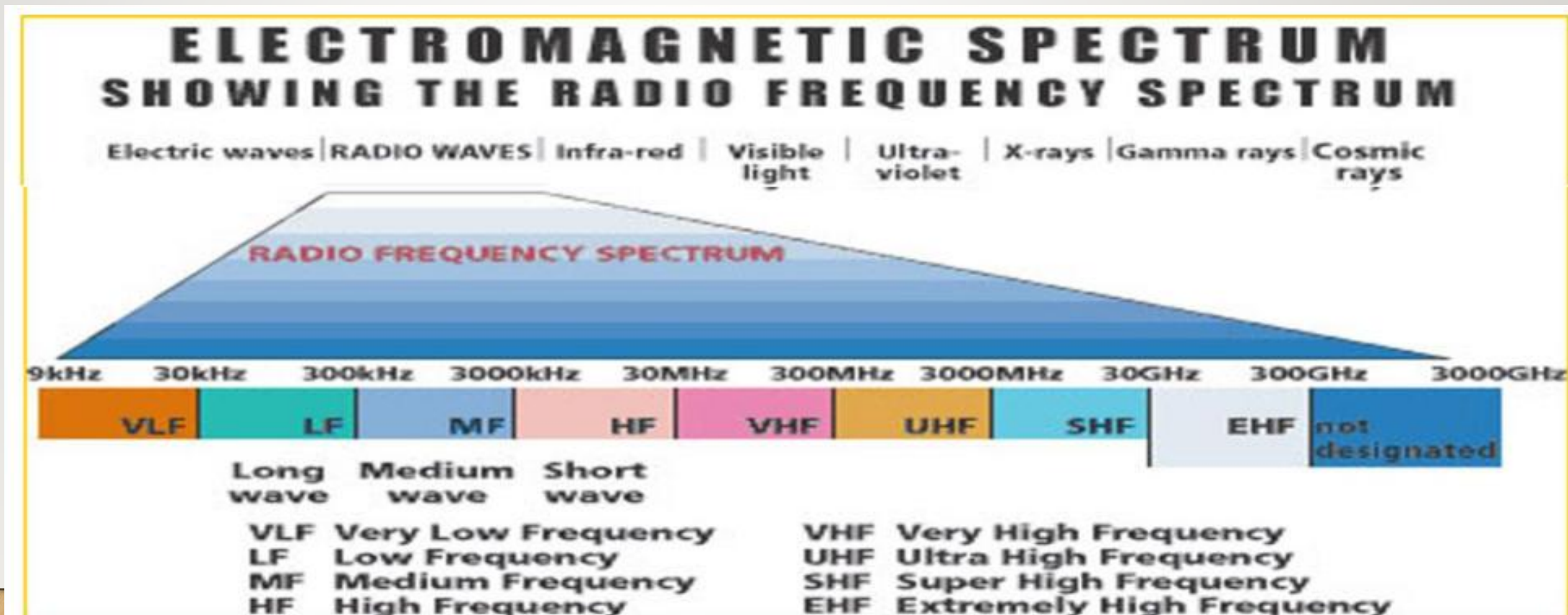
- Ways of administering Spectrum
- Spectrum Cost and Cellular networks
- Spectrum Congestion and its effects
- TV white space

Ways of Administering Spectrum

- Spectrum management (administering) regulates the allocation, assignment, and use of radio frequency spectrum to ensure efficient and effective utilization of this finite resource.
- Spectrum is the portion of electromagnetic waves between 9 kHz to 3000 GHz,

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➤ Radio Frequency Spectrum in the Electromagnetic Spectrum





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Spectrum is allocated to broadcasting, mobile, and satellite services under ITU's Radio Regulations (RR).

Key consumer uses of the spectrum include:

- Browsing the internet via mobile networks or Wi-Fi.
- Making phone calls or sending text messages over mobile networks.
- Watching television or listening to the radio.
- Using Bluetooth to connect personal and home devices.
- Accessing navigation services on smartphones or in cars.



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- Spectrum management balances competing demands among users while maximizing utilization and fostering wireless innovation.
- Spectrum Administering Process

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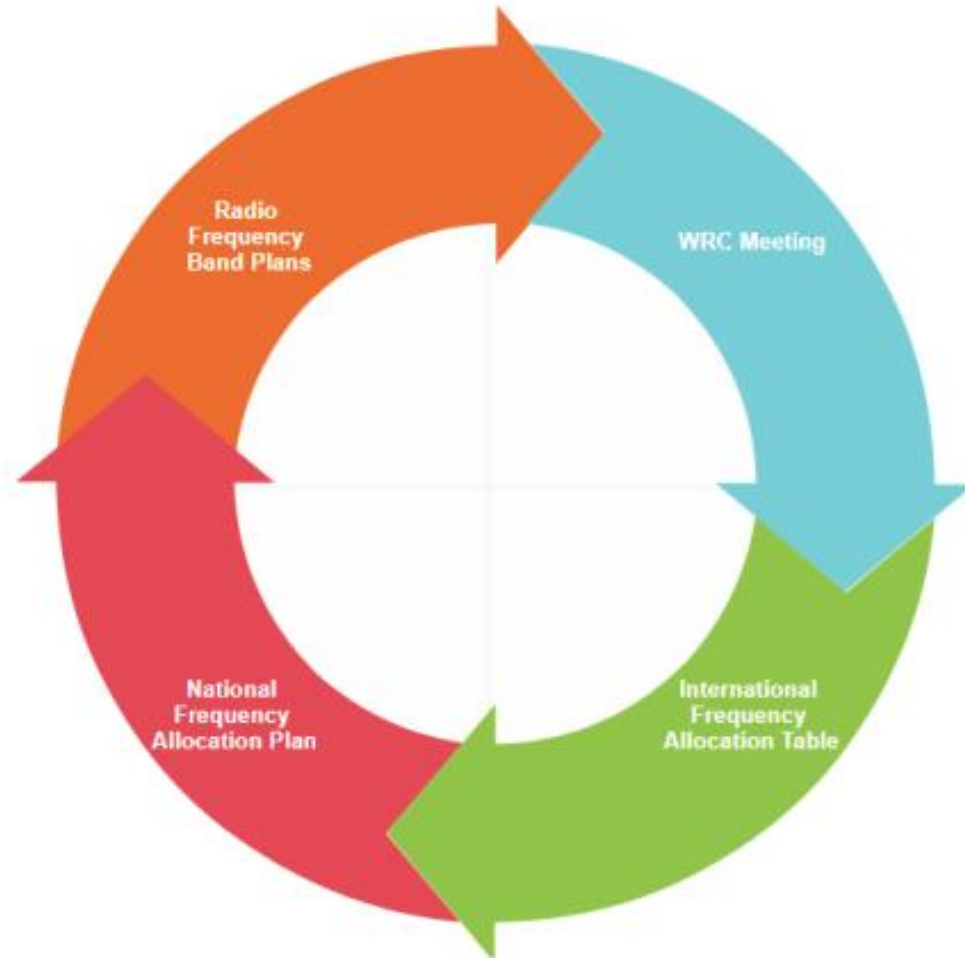




Spectrum management process.

➤ Radio Frequency Planning and Allocation Process

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➤ Spectrum planning and allocation,

Coordinated by the ITU, it involves activities spanning from the international to the national level.

- The ITU's World Radiocommunication Conferences (WRCs) review and revise the Radio Regulations, setting the global framework for spectrum allocation and satellite orbit use.
- After each WRC, the ITU updates the International Frequency Allocation Table (IFAT) in Article 5 of the Radio Regulations, guiding national regulatory authorities.

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- i. The Tanzania Communications Regulatory Authority (TCRA) develops and updates the National Frequency Allocation Plan (NFAP) based on IFAT, considering the country's needs, infrastructure, and future communication requirements.
- ii. TCRA develops radio frequency band plans based on the NFAP, allocating spectrum to mobile, broadcasting, aviation, and emergency services, ensuring efficient use and minimizing interference.

Spectrum licensing and assignment

- Spectrum licensing and assignment is the process of assigning frequency channels to service providers to enable them to offer services.
- Spectrum licensing and assignment, overseen by TCRA, involves three methods:
 - Administrative method: Allocates channels in a "first come, first served" manner when availability exceeds demand.

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- Market-based method: Uses competitive auctions when demand exceeds supply, ensuring efficient allocation.
- Commons or license-exempt: Treats spectrum as a shared resource, promoting innovation and coordination in unlicensed bands.

Spectrum Monitoring and Enforcement

Spectrum monitoring is crucial for effective spectrum management. TCRA conducts spectrum monitoring to fulfill important functions such as;

- Spectrum management involves overseeing the allocation, assignment, and use of radio frequencies to ensure their efficient, fair, and effective utilization.
- Spectrum monitoring enables assess spectrum utilization across different frequency bands and geographical areas. By analyzing usage patterns and demand trends, the authority gains valuable insights into how spectrum resources are being utilized



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- Spectrum monitoring provides essential feedback for improving spectrum planning and assignment processes. By identifying areas of congestion, underutilization, or emerging trends for adjusting strategies and policies to better meet the evolving needs of stakeholders and ensure efficient spectrum management.



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- The spectrum monitoring is done through the use of equipment that adheres to international standards, ensuring accuracy, reliability, and compatibility with global best practices. TCRA can effectively fulfill its role as the "eyes and ears" of the spectrum management process.

Spectrum Cost and Cellular Networks

- Spectrum, the airwaves used for wireless signals, is limited. This scarcity drives up costs for cellular companies.
- High Prices: Companies pay hefty fees to governments for the right to use specific spectrum bands.
- Limited Supply: High demand for mobile service means fierce competition for this limited resource.
- Impact: These costs increase prices for consumers and can slow down the development of faster, more reliable networks.

Spectrum Congestion and its Effects

- Spectrum Congestion: This happens when too many devices try to use the same radio frequencies (like the "lanes" on the highway).
- Slow Speeds: This "traffic jam" slows down your internet connection, making things like video calls and streaming laggy.
- Essentially, when there's too much demand on the available airwaves, it leads to slower and less reliable wireless service.

TV White space

- TV white space is a bit like finding empty parking spots on a busy street. It refers to unused parts of the radio spectrum that were originally set aside for TV broadcasts but aren't always used.
- Scarcity of Spectrum: Just like parking spaces, the radio spectrum is a limited resource.
- Finding the "Empty Spots": TV white space is like discovering those empty parking spots.



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- Using the "Empty Spots": This "empty space" can be used for other wireless services, like bringing internet to rural areas or creating new wireless networks.
- Essentially, TV white space helps us make better use of the limited radio spectrum by utilizing those unused frequencies.



Thank you

➤ <https://tcra.go.tz/uploads/documents/sw-1727101237-Spectrum%20Management%20Strategy%20-%202024:2025-2034:2035.pdf>